ALGEBRA 2 · POLYNOMIAL FUNCTIONS

Zeros & Multiplicity Builder

Explore & Reflect

Name _____ Date _____ Period _____

Open **Zeros & Multiplicity Builder** on Mr. K's Classroom: Algebra 2 → Polynomial Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. What do you think the graph does at a zero that appears twice, like $(x - 2)^2$?

2 Explore use the tool and record what you see

Build each polynomial

Function	Degree	End behavior (left, right)	At each zero
$f(x) = (x + 3)(x - 1)(x - 4)$			
$f(x) = (x - 2)^2(x + 1)$			
$f(x) = -(x + 1)^3$			
$f(x) = x^2(x - 3)^2$			
$f(x) = -(x - 1)(x + 2)^2$			

3 Write the rule fill in each blank from what you observed

1. At a zero with odd multiplicity the graph _____ the x-axis. With even multiplicity it _____.
2. Multiplicity 3 or more makes the graph _____ near the zero.
3. The degree is the _____ of the multiplicities.
4. Even degree: both ends point the _____ way. Odd degree: the ends point _____ ways.
5. A negative leading coefficient makes the right end point _____.

4 Check no tool: show your work

1. Describe $f(x) = (x - 5)^2(x + 2)^3$.

2. Write a polynomial with leading coefficient 1, a zero at −1 of multiplicity 2 and a zero at 4.

3. Describe the end behavior of $f(x) = -2x^4 + x$.

5 Reflect

Why does an even multiplicity make the graph bounce?